

Baseline: Agoda vs Booking Aggregate Scores (N=10 hotels)**The MEANS Procedure**

Variable	Mean	Std Dev	Minimum	Maximum
agoda_agg_score	9.0200000	0.1475730	8.8000000	9.2000000
booking_agg_score	9.3900000	0.1852926	9.2000000	9.7000000
score_gap	-0.3700000	0.1059350	-0.5000000	-0.2000000

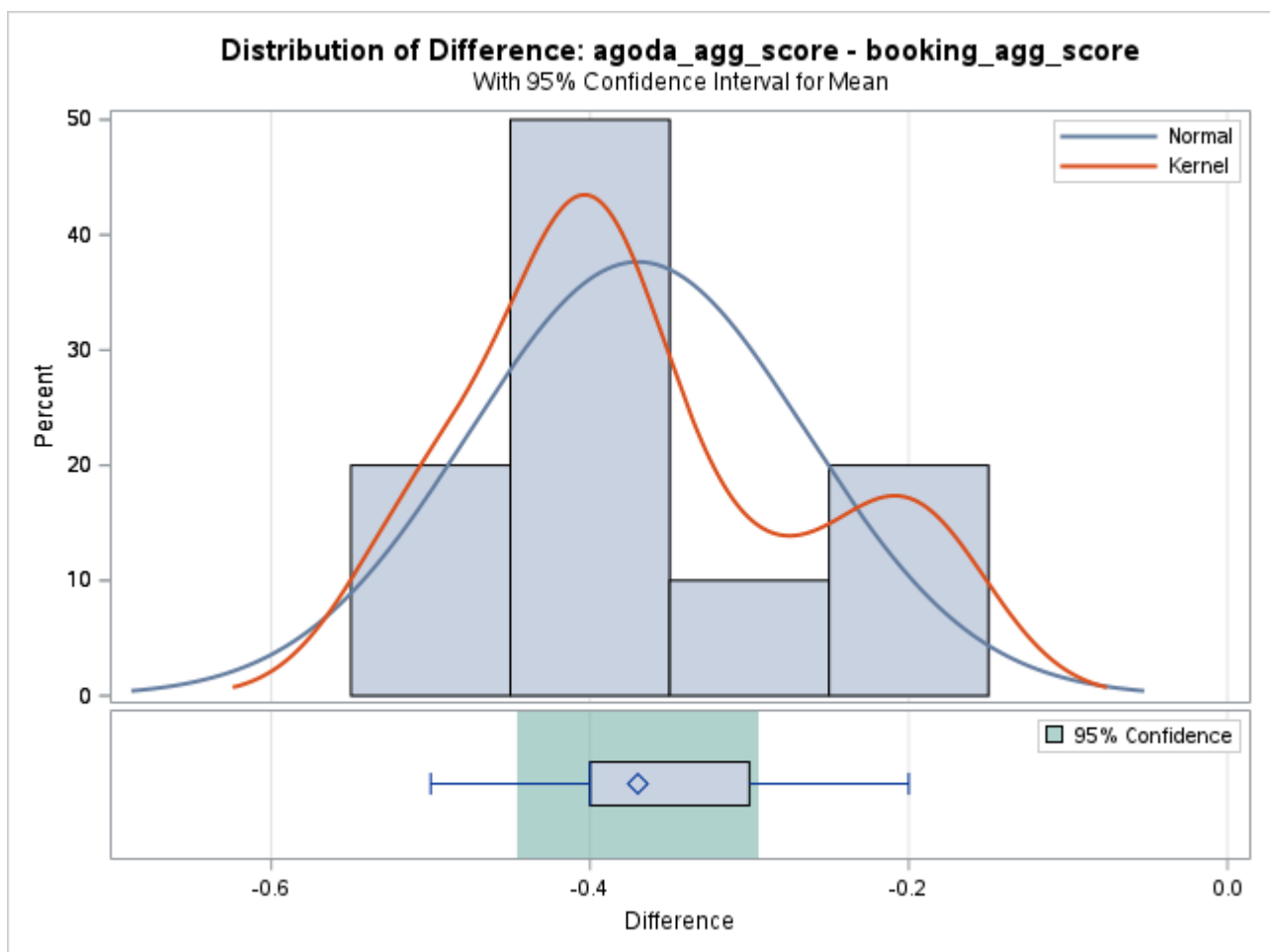
Paired T-Test: Agoda vs Booking Score Gap (N=10, DF=9)**The TTEST Procedure**

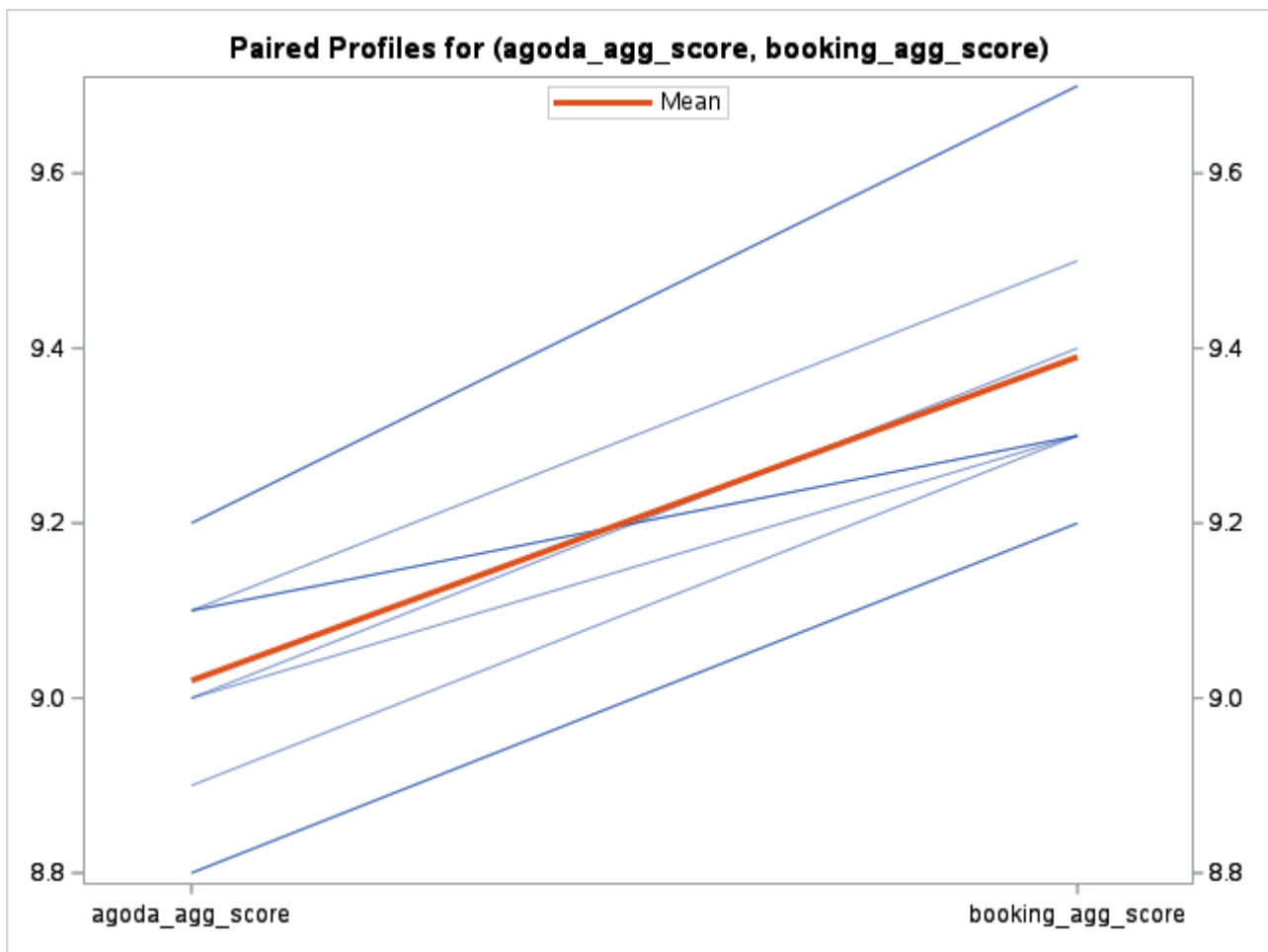
Difference: agoda_agg_score - booking_agg_score

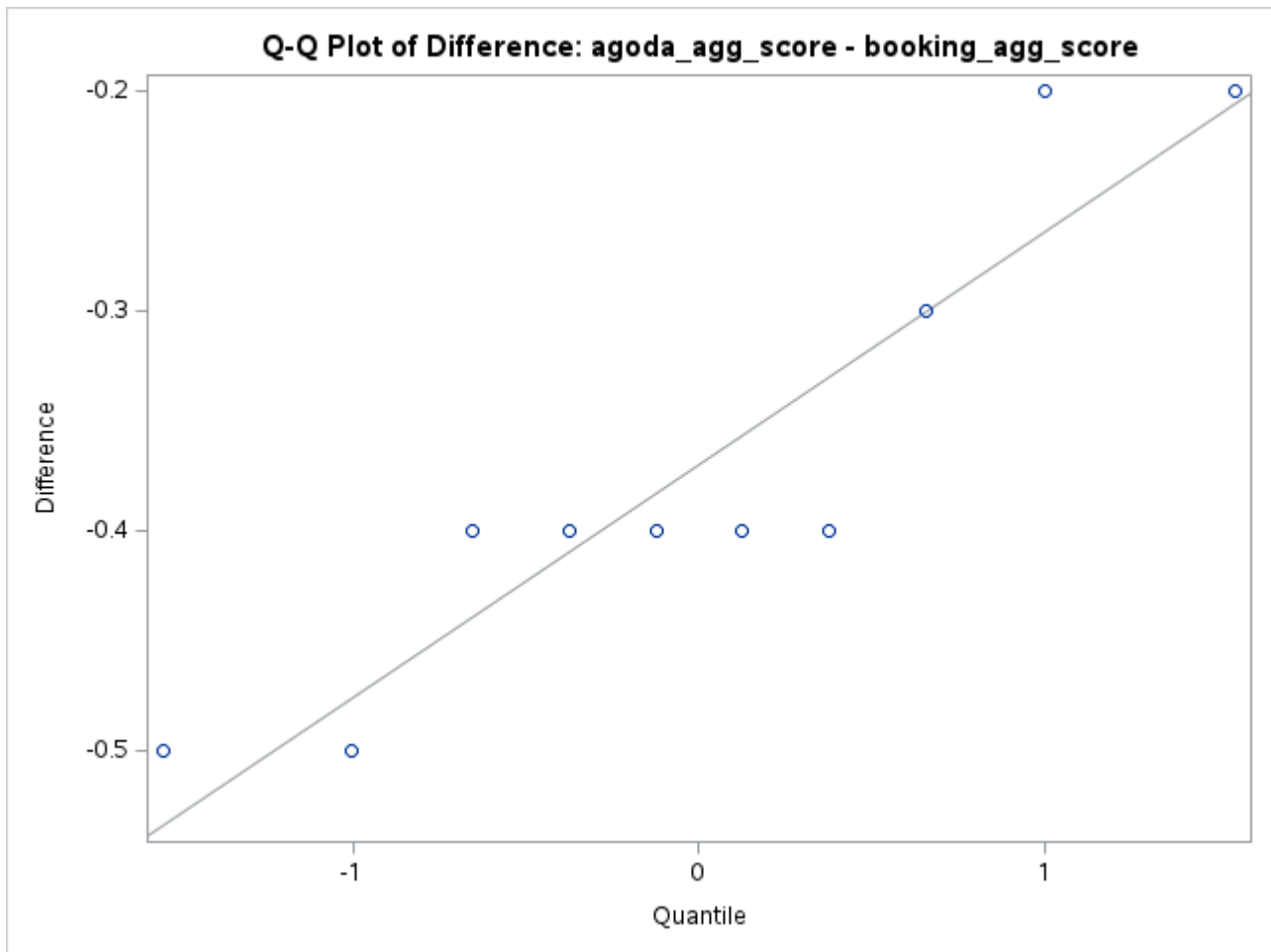
N	Mean	Std Dev	Std Err	Minimum	Maximum
10	-0.3700	0.1059	0.0335	-0.5000	-0.2000

Mean	95% CL Mean	Std Dev	95% CL Std Dev
-0.3700	-0.4458 -0.2942	0.1059	0.0729 0.1934

DF	t Value	Pr > t
9	-11.04	<.0001







Sub-Score Gaps by Category (Agoda minus Booking), averaged across 10 hotels

The MEANS Procedure

Analysis Variable : gap		
category	N Obs	Mean
cleanliness	10	-0.1400000
facilities	10	-0.2300000
location	10	-0.3700000
roomComfort	10	-0.4100000
staffPerformance	10	-0.3300000
valueForMoney	10	-0.1875000

H1 Model 1a: Raw platform effect on score (naive)

The REG Procedure
Model: MODEL1
Dependent Variable: score

Number of Observations Read	1851
Number of Observations Used	1851

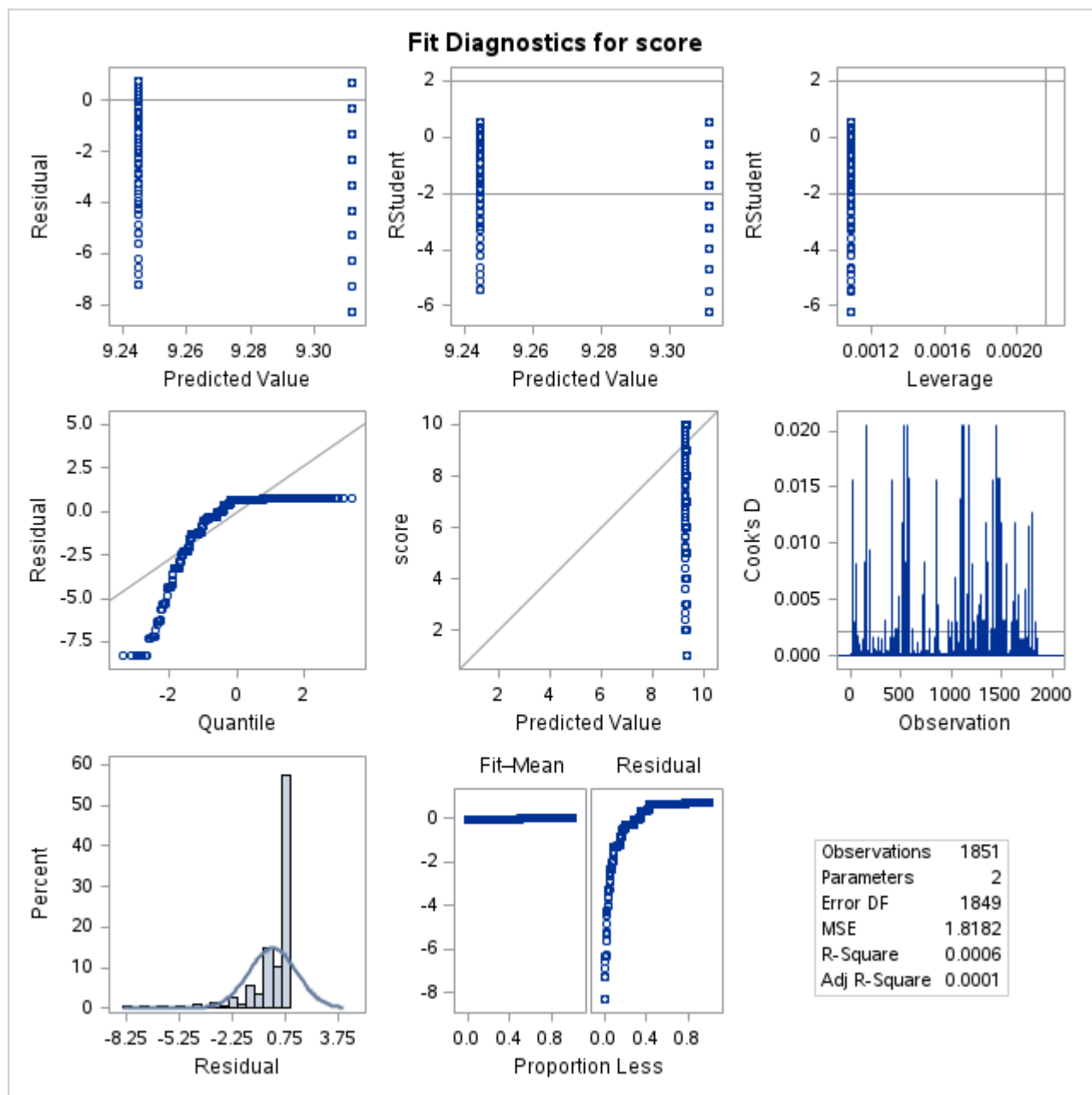
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	2.07438	2.07438	1.14	0.2856
Error	1849	3361.84690	1.81820		
Corrected Total	1850	3363.92127			

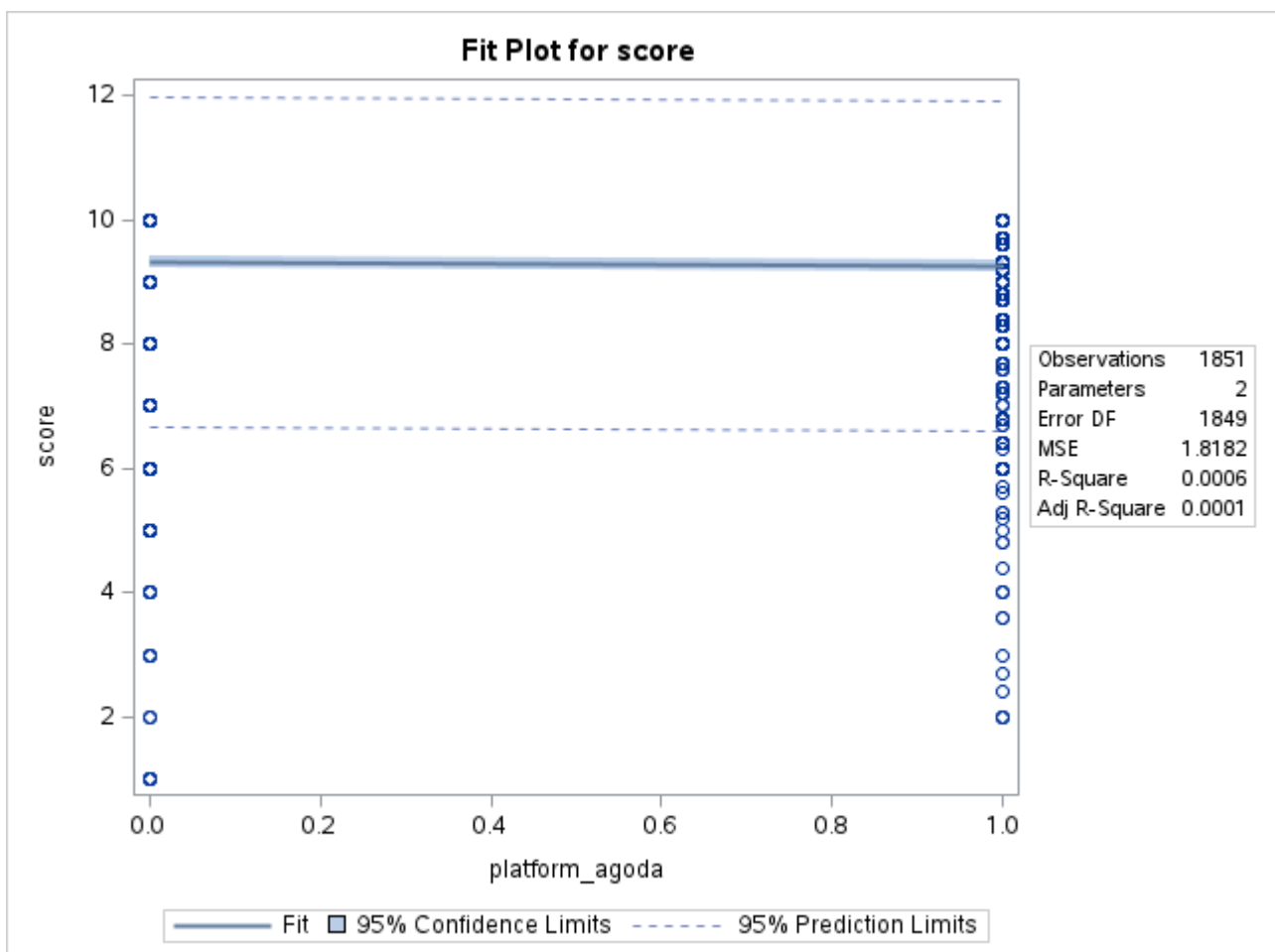
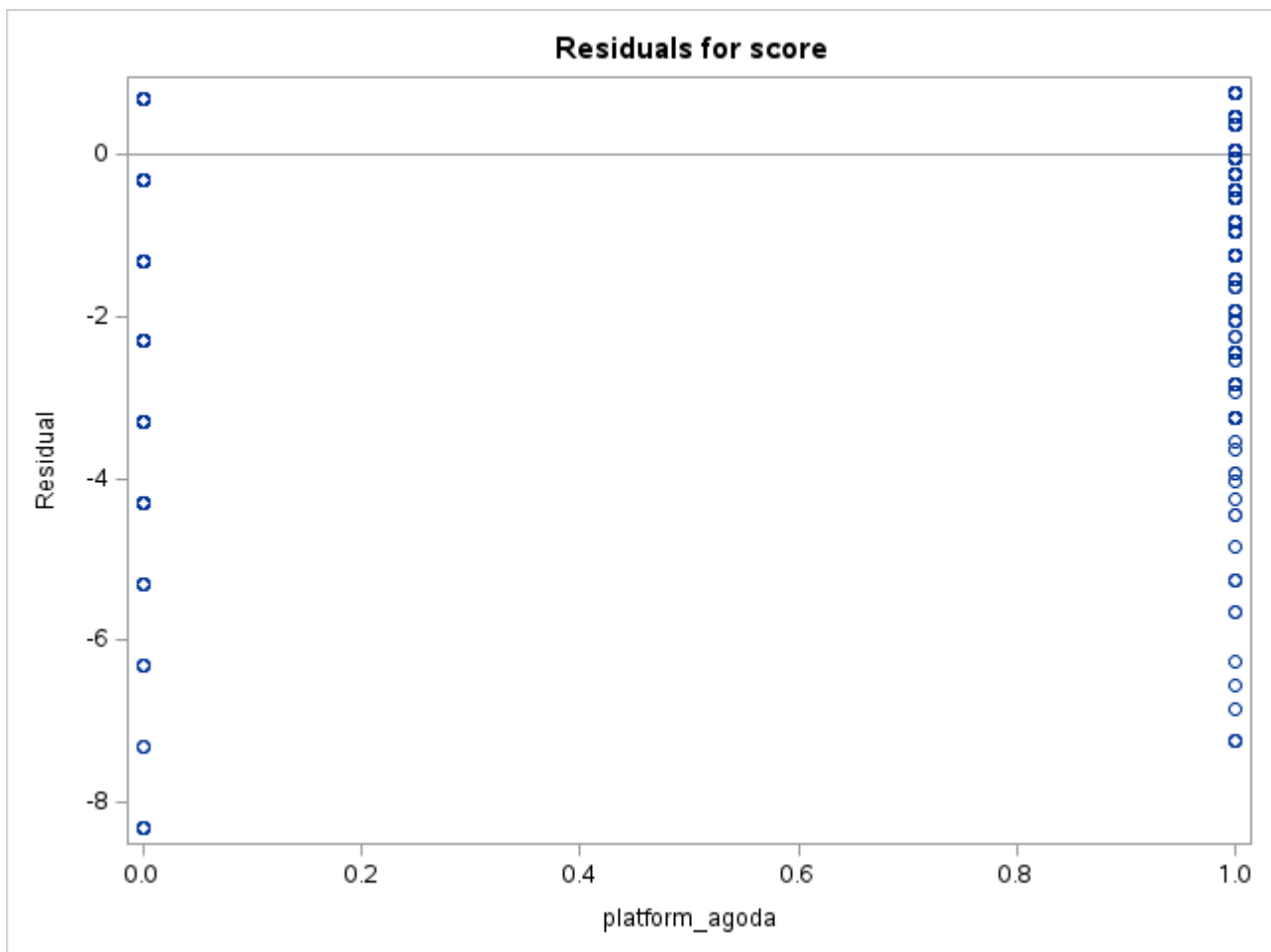
Root MSE	1.34841	R-Square	0.0006
Dependent Mean	9.27834	Adj R-Sq	0.0001
Coeff Var	14.53284		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	9.31176	0.04429	210.26	<.0001
platform_agoda	1	-0.06695	0.06268	-1.07	0.2856

H1 Model 1a: Raw platform effect on score (naive)

The REG Procedure
Model: MODEL1
Dependent Variable: score





H1 Model 1b: Platform effect controlling for hotel fixed effects**The GLM Procedure**

Class Level Information		
Class	Levels	Values
hotel_id	10	4season_bali_atSayan Park_Hayatt alila_villas_uluwatu mandarin_oriental_si mandarin_oriental_to marina_bay_sands raffles_singapore ritz_carlton st_regis_bali_resort the_peninsula

Number of Observations Read	1851
Number of Observations Used	1851

H1 Model 1b: Platform effect controlling for hotel fixed effects**The GLM Procedure**

Dependent Variable: score

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	10	108.369302	10.836930	6.12	<.0001
Error	1840	3255.551973	1.769322		
Corrected Total	1850	3363.921275			

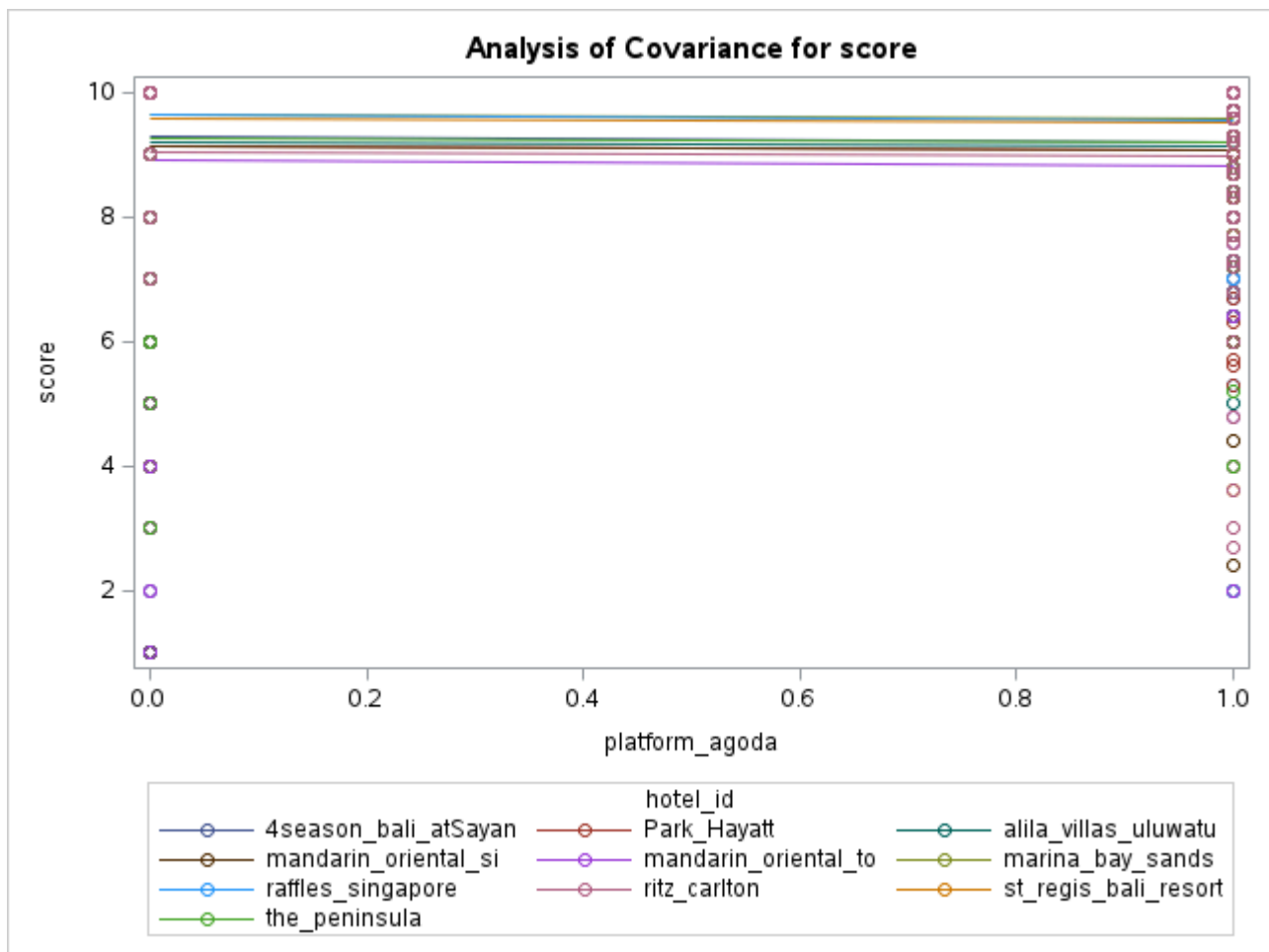
R-Square	Coeff Var	Root MSE	score Mean
0.032215	14.33618	1.330159	9.278336

Source	DF	Type I SS	Mean Square	F Value	Pr > F
platform_agoda	1	2.0743762	2.0743762	1.17	0.2790
hotel_id	9	106.2949254	11.8105473	6.68	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
platform_agoda	1	2.3084659	2.3084659	1.30	0.2535
hotel_id	9	106.2949254	11.8105473	6.68	<.0001

Parameter	Estimate		Standard Error	t Value	Pr > t
Intercept	9.267928139	B	0.09917643	93.45	<.0001
platform_agoda	-0.071856278		0.06290807	-1.14	0.2535
hotel_id 4season_bali_atSayan	0.022384205	B	0.13570930	0.16	0.8690
hotel_id Park_Hayatt	-0.114000000	B	0.13301585	-0.86	0.3915
hotel_id alila_villas_uluwatu	-0.049000000	B	0.13301585	-0.37	0.7126
hotel_id mandarin_oriental_si	-0.121000000	B	0.13301585	-0.91	0.3631
hotel_id mandarin_oriental_to	-0.358209118	B	0.14620766	-2.45	0.0144
hotel_id marina_bay_sands	0.396000000	B	0.13301585	2.98	0.0029
hotel_id raffles_singapore	0.372500000	B	0.13301585	2.80	0.0052
hotel_id ritz_carlton	-0.227292345	B	0.15343625	-1.48	0.1387
hotel_id st_regis_bali_resort	0.319000000	B	0.13301585	2.40	0.0166
hotel_id the_peninsula	0.000000000	B	.	.	.

Note: The X'X matrix has been found to be singular, and a generalized inverse was used to solve the normal equations. Terms whose estimates are followed by the letter 'B' are not uniquely estimable.



H1 Model 2: Nationality effect controlling for platform + hotel FE (cross-platform)

The GLM Procedure

Class Level Information		
Class	Levels	Values
hotel_id	10	4season_bali_atSayan Park_Hayatt alila_villas_uluwatu mandarin_oriental_si mandarin_oriental_to marina_bay_sands raffles_singapore ritz_carlton st_regis_bali_resort the_peninsula

Number of Observations Read	1850
Number of Observations Used	1850

H1 Model 2: Nationality effect controlling for platform + hotel FE (cross-platform)

The GLM Procedure

Dependent Variable: score

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	11	117.154733	10.650430	6.03	<.0001
Error	1838	3246.663019	1.766411		
Corrected Total	1849	3363.817751			

R-Square	Coeff Var	Root MSE	score Mean
0.034828	14.32465	1.329064	9.278162

Source	DF	Type I SS	Mean Square	F Value	Pr > F
asian_reviewer	1	14.1277012	14.1277012	8.00	0.0047

Source	DF	Type I SS	Mean Square	F Value	Pr > F
platform_agoda	1	0.3341258	0.3341258	0.19	0.6637
hotel_id	9	102.6929056	11.4103228	6.46	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
asian_reviewer	1	8.6826639	8.6826639	4.92	0.0267
platform_agoda	1	0.0338664	0.0338664	0.02	0.8899
hotel_id	9	102.6929056	11.4103228	6.46	<.0001

Parameter	Estimate		Standard Error	t Value	Pr > t
Intercept	9.300862306	B	0.10018784	92.83	<.0001
asian_reviewer	-0.166112682		0.07492421	-2.22	0.0267
platform_agoda	0.010115675		0.07305601	0.14	0.8899
hotel_id 4season_bali_atSayan	0.012844883	B	0.13566558	0.09	0.9246
hotel_id Park_Hayatt	-0.138916902	B	0.13338071	-1.04	0.2978
hotel_id alila_villas_uluwatu	-0.068228082	B	0.13329280	-0.51	0.6088
hotel_id mandarin_oriental_si	-0.115186056	B	0.13293226	-0.87	0.3863
hotel_id mandarin_oriental_to	-0.360034091	B	0.14608971	-2.46	0.0138
hotel_id marina_bay_sands	0.381049859	B	0.13307734	2.86	0.0042
hotel_id raffles_singapore	0.350074788	B	0.13329072	2.63	0.0087
hotel_id ritz_carlton	-0.208060795	B	0.15355212	-1.35	0.1756
hotel_id st_regis_bali_resort	0.323152817	B	0.13291959	2.43	0.0151
hotel_id the_peninsula	0.000000000	B	.	.	.

Note: The X'X matrix has been found to be singular, and a generalized inverse was used to solve the normal equations. Terms whose estimates are followed by the letter 'B' are not uniquely estimable.

H1: Reviewer Profile by Platform (N=10 hotels)

The MEANS Procedure

platform	N Obs	Variable	Mean
Agoda	10	pct_asian	65.1200000
		pct_solo	10.7300000
		pct_couple	50.8200000
		pct_business	6.8800000
		avg_score	9.2280000
		pct_nationality_known	99.9000000
Booking.com	10	pct_asian	16.9400000
		pct_solo	12.0100000
		pct_couple	48.9000000
		pct_business	0
		avg_score	9.2860000
		pct_nationality_known	100.0000000

H2: Score Variance Comparison (Agoda vs Booking), N=10 hotels

The TTEST Procedure

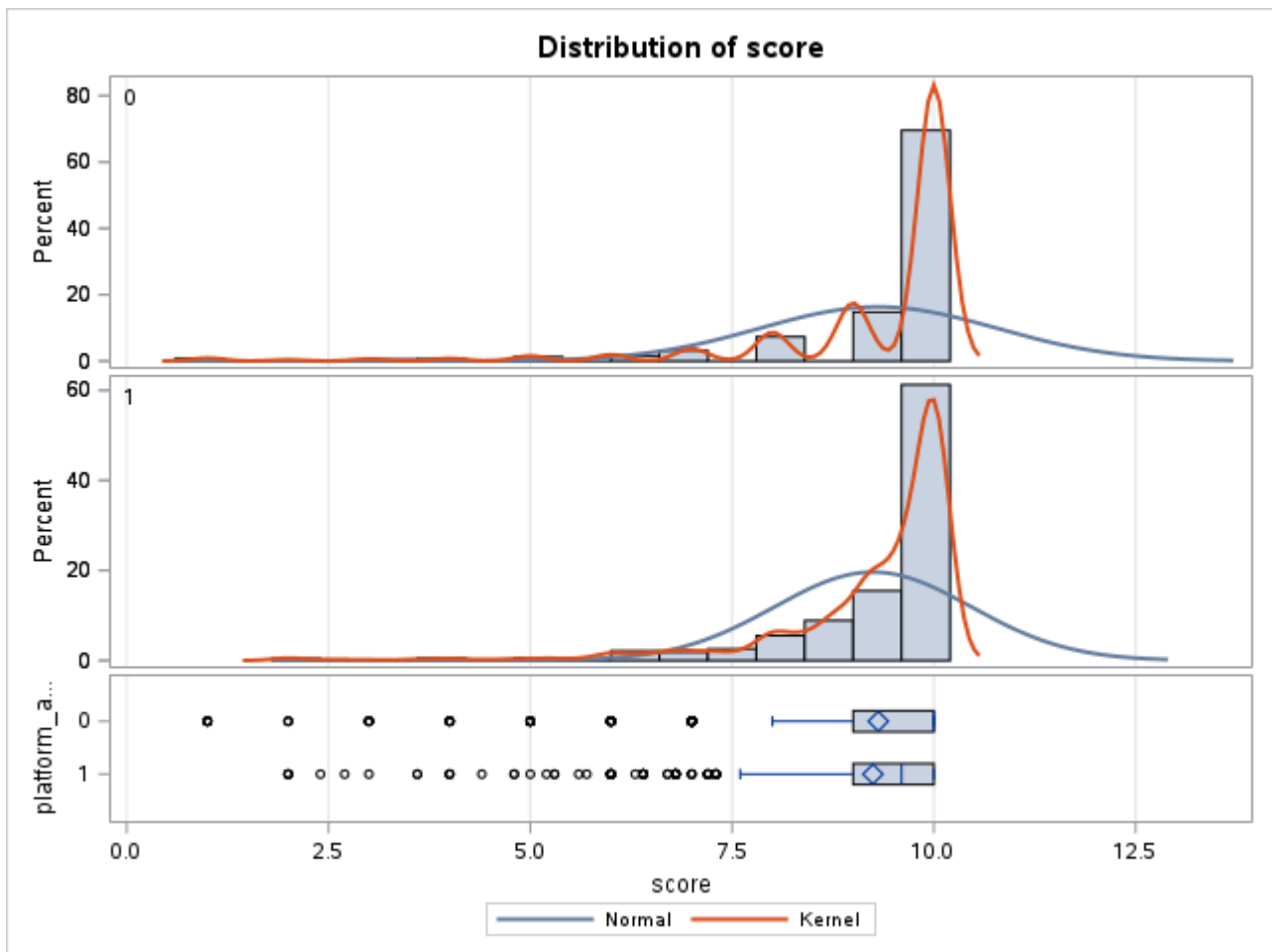
Variable: score

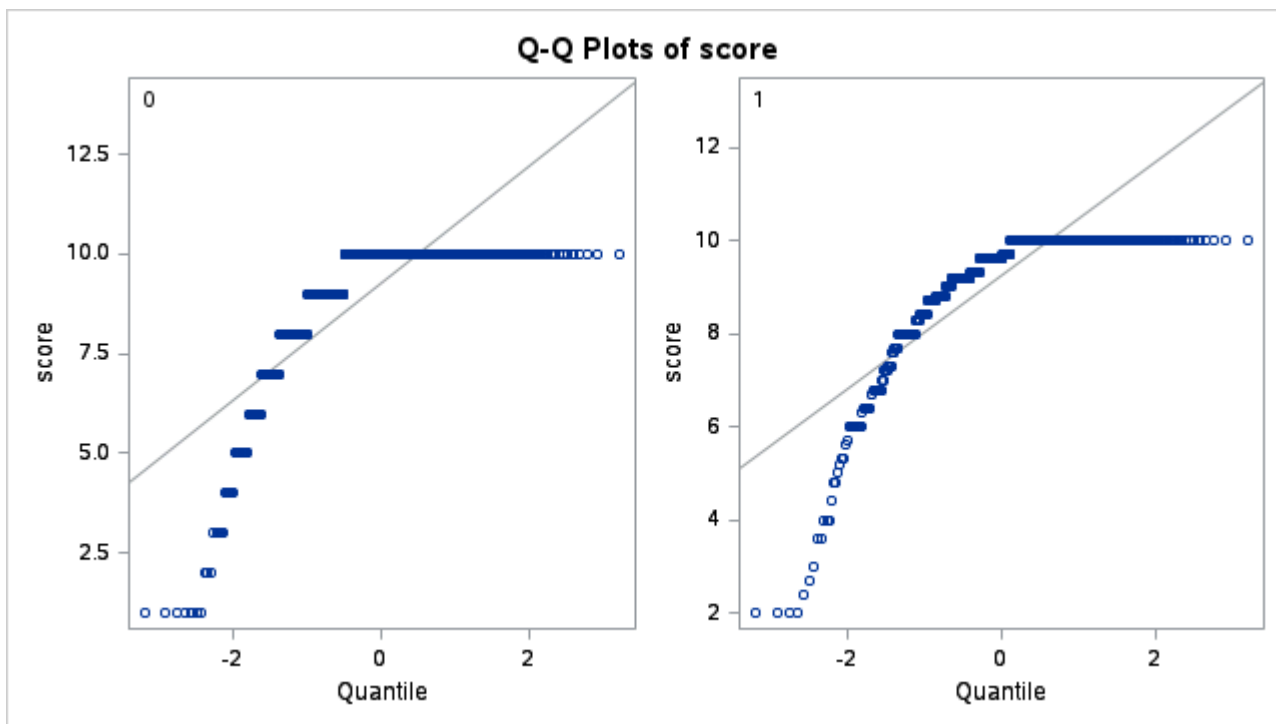
platform_agoda	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		927	9.3118	1.4663	0.0482	1.0000	10.0000
1		924	9.2448	1.2187	0.0401	2.0000	10.0000
Diff (1-2)	Pooled		0.0670	1.3484	0.0627		
Diff (1-2)	Satterthwaite		0.0670		0.0627		

platform_agoda	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		9.3118	9.2172	9.4063	1.4663	1.4024	1.5363
1		9.2448	9.1661	9.3235	1.2187	1.1656	1.2770
Diff (1-2)	Pooled	0.0670	-0.0560	0.1899	1.3484	1.3063	1.3933
Diff (1-2)	Satterthwaite	0.0670	-0.0559	0.1899			

Method	Variances	DF	t Value	Pr > t
Pooled	Equal	1849	1.07	0.2856
Satterthwaite	Unequal	1791.2	1.07	0.2855

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	926	923	1.45	<.0001





H2: Score Distribution Statistics by Platform

The MEANS Procedure

Analysis Variable : score						
platform	N Obs	Mean	Std Dev	Variance	Minimum	Maximum
Agoda	924	9.2448052	1.2187347	1.4853143	2.0000000	10.0000000
Booki	927	9.3117584	1.4662885	2.1500020	1.0000000	10.0000000

BA: Hotel Risk Segmentation + Revenue at Risk (N=10)

Obs	hotel_name	city	score_gap	risk_tier	annual_revenue_at_risk
1	Four Seasons Resort Bali at Sayan	Bali	-0.4	At-Risk	3456
2	The St. Regis Bali Resort	Bali	-0.5	At-Risk	3456
3	Alila Villas Uluwatu	Bali	-0.4	At-Risk	3456
4	Marina Bay Sands	Singapore	-0.4	At-Risk	3456
5	Raffles Singapore	Singapore	-0.5	At-Risk	3456
6	Mandarin Oriental Singapore	Singapore	-0.3	Watch	3456
7	Park Hyatt Tokyo	Tokyo	-0.4	At-Risk	3456
8	Mandarin Oriental Tokyo	Tokyo	-0.2	Watch	3456
9	The Peninsula Tokyo	Tokyo	-0.2	Watch	3456
10	The Ritz-Carlton Tokyo	Tokyo	-0.4	At-Risk	3456

BA: Total Annual Revenue at Risk by Tier

The MEANS Procedure

Analysis Variable : annual_revenue_at_risk			
risk_tier	N Obs	Sum	Mean
At-Risk	7	24192.00	3456.00
Watch	3	10368.00	3456.00